Grokking Data Structures & Algorithms for Coding Interviews

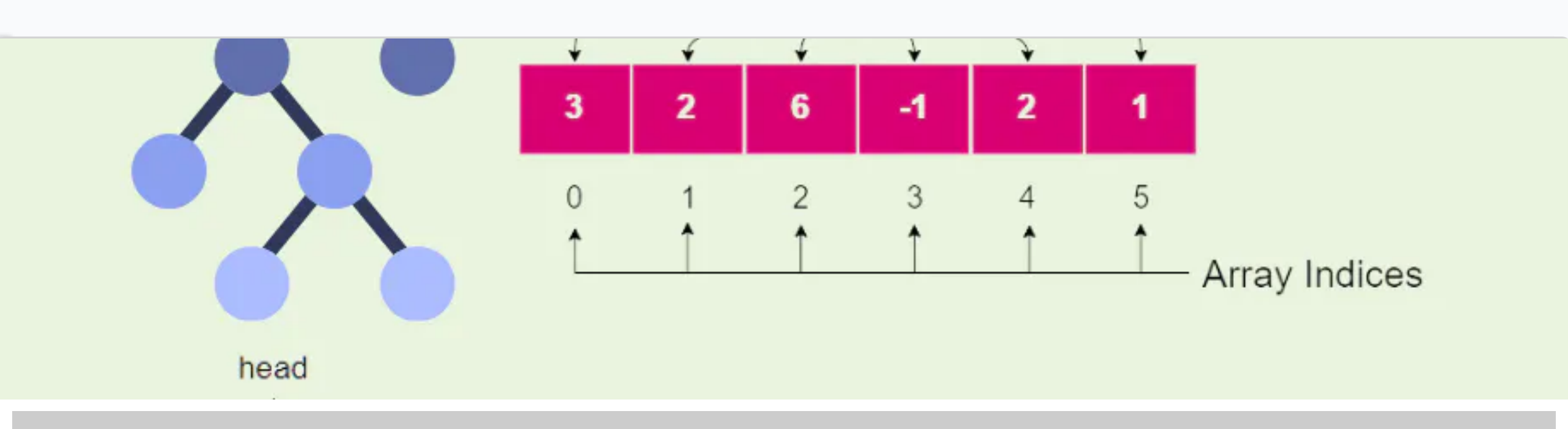
Unlock Coding Interview Success: Dive Deep into Data Structures and Algorithms!

Level: Beginner | Study Time: 50h | Lessons: 165 | Playgrounds : 185

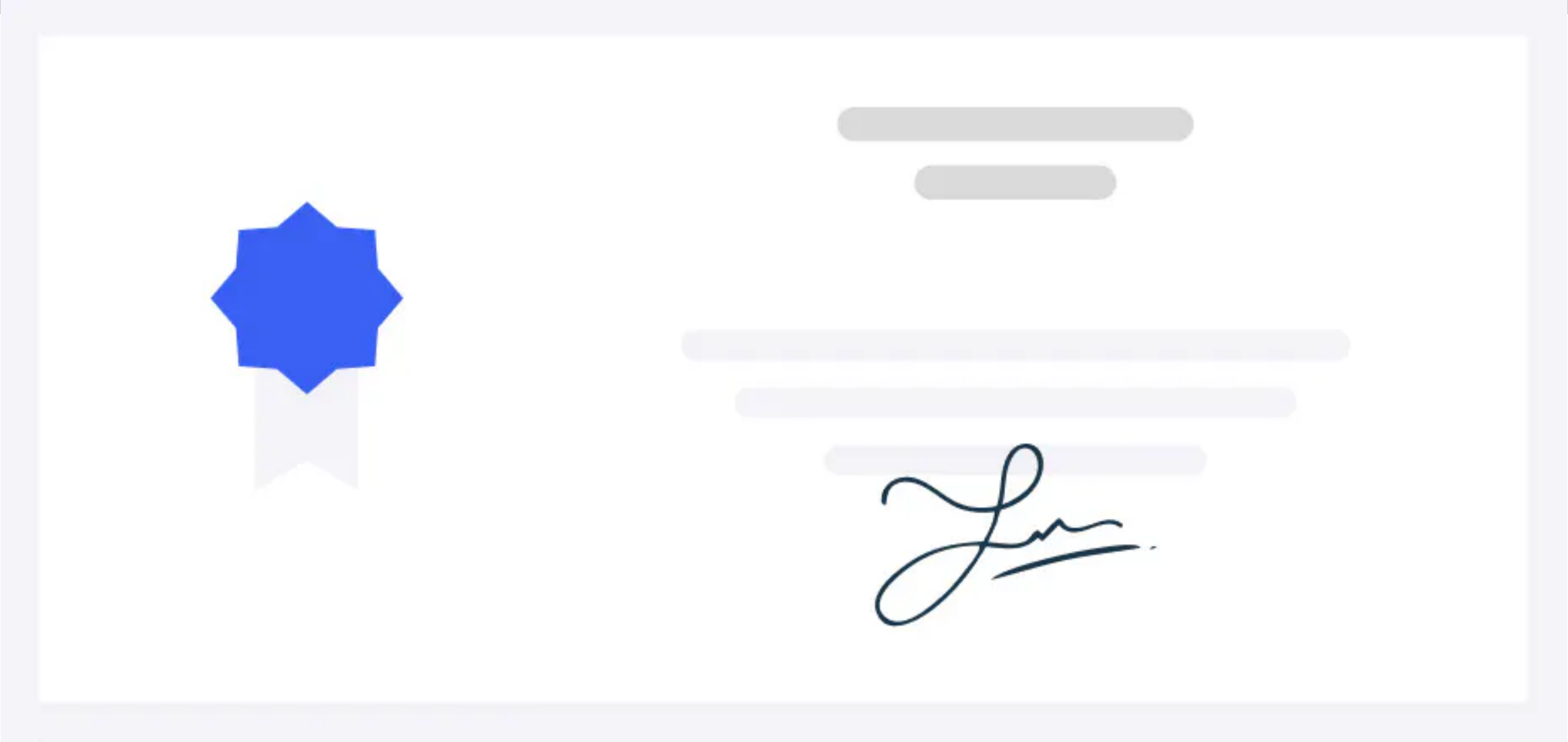
Course Overview

In this comprehensive course, we delve deep into the world of data structures & algorithms, covering both the commonly used and the more advanced topics. Participants will learn the principles, usage, and implementation of various data structures, along with their time and space complexities. The course will feature multiple coding problems for each data structure and is tailored for coding interviews.

By the end of this course, learners will be adept at selecting and applying the most suitable data structure for any coding problem, ensuring optimized code performance. Prepare to elevate your coding skills and ace those coding interviews with confidence!



Start Reading



Certification on completing the course.

Download Certificate

Course Content

1. Introduction to Data Structures
4 chapters

- Who Should Take This Course?
- Understanding Data Structures
- Types of Data Structures
- An Overview of Big-O

2. Array
10 chapters

- Introduction to Arrays
- Arrays in Different Programming Languages
- Problem 1: Running Sum of 1d Array (easy)
- Solution: Running Sum of 1d Array (easy)
- Problem 2: Contains Duplicate (easy)
- Solution: Contains Duplicate
- Problem 3: Left and Right Sum Differences (easy)
- Solution: Left and Right Sum Differences (easy)
- Problem 4: Find the Highest Altitude(easy)
- Solution: Find the Highest Altitude (easy)

3. Matrix
7 chapters

- Introduction to Matrix
- Problem 1: Richest Customer Wealth (easy)
- Solution: Richest Customer Wealth (easy)
- Problem 2: Matrix Diagonal Sum (easy)
- Solution: Matrix Diagonal Sum (easy)
- Problem 3: Row With Maximum Ones(easy)
- Solution: Row With Maximum Ones(easy)

4. Stack
22 chapters

- Introduction to Stack
- Operations on Stack
- Implementing Stack Data Structure
- Applications of Stack
- Problem 1: Balanced Parentheses
- Solution: Balanced Parentheses
- Problem 2: Reverse a String
- Solution: Reverse a String
- Problem 3: Decimal to Binary Conversion
- Solution: Decimal to Binary Conversion
- Problem 4: Next Greater Element
- Solution: Next Greater Element
- Problem 5: Sorting a Stack
- Solution: Sorting a Stack
- Problem 6: Simplify Path
- Solution: Simplify Path
- Problem 7: Remove All Adjacent Duplicates In String
- Solution: Remove All Adjacent Duplicates In String
- Problem 8: Removing Stars From a String
- Solution: Removing Stars From a String
- Problem 9: Make The String Great
- Solution: Make The String Great

5. Queue
16 chapters

- Introduction to Queues
- Working with Simple Queues
- Diving Deeper – Circular Queues and Deques
- Applications and Advanced Concepts
- Problem 1: Reverse a Queue
- Solution: Reverse a Queue
- Problem 2: Implement Stack using Queues
- Solution: Implement Stack using Queues
- Problem 3: Generate Binary Numbers from 1 to N
- Solution: Generate Binary Numbers from 1 to N
- Problem 4: Palindrome Check using Queue
- Solution: Palindrome Check using Queue
- Problem 5: Zigzag Iterator
- Solution: Zigzag Iterator
- Problem 6: Max of All Subarrays of Size 'k'
- Solution: Max of All Subarrays of Size 'k'

6. LinkedList
13 chapters

- Introduction to LinkedList
- Operations on Singly Linked List
- Operations on Doubly Linked List
- Problem 1: Reverse Linked List (easy)
- Solution: Reverse Linked List (easy)
- Problem 2: Remove Duplicates from Sorted List (easy)
- Solution: Remove Duplicates from Sorted List (easy)
- Problem 3: Merge Two Sorted Lists (easy)
- Solution: Merge Two Sorted Lists (easy)
- Problem 4: Find if Doubly Linked List is a Palindrome (easy)
- Solution: Find if Doubly Linked List is a Palindrome (easy)
- Problem 5: Swap Nodes in Pairs (medium)
- Solution: Swap Nodes in Pairs (easy)

7. Tree & Binary Search Tree
16 chapters

- Introduction to Tree
- Binary Search Tree (BST)
- Maximum Depth (or Height) of Binary Tree (easy)
- Solution: Maximum Depth (or Height) of Binary Tree (easy)
- Balanced Binary Tree (easy)
- Solution: Balanced Binary Tree (easy)
- Minimum Difference Between BST Nodes (easy)
- Solution: Minimum Difference Between BST Nodes (easy)
- Range Sum of BST (easy)
- Solution: Range Sum of BST (easy)
- Kth Smallest Element in a BST (medium)
- Solution: Kth Smallest Element in a BST (medium)
- Closest Binary Search Tree Value (medium)
- Solution: Closest Binary Search Tree Value (medium)
- Merge Two Binary Trees (medium)
- Solution: Merge Two Binary Trees (medium)

8. Hash Table (aka Hashmap or Dictionary)
13 chapters

- Introduction to Hashing
- Introduction to Hash Tables
- Issues with Hash Tables
- Problem 1: First Non-repeating Character (easy)
- Solution: First Non-repeating Character (easy)
- Problem 2: Largest Unique Number (easy)
- Solution: Largest Unique Number (easy)
- Problem 3: Maximum Number of Balloons (easy)
- Solution: Maximum Number of Balloons (easy)
- Problem 4: Longest Palindrome(easy)
- Solution: Longest Palindrome(easy)
- Problem 5: Ransom Note (easy)
- Solution: Ransom Note (easy)

9. HashSet
9 chapters

- Introduction to HashSets
- Problem 1: Counting Elements (easy)
- Solution: Counting Elements (easy)
- Problem 2: Jewels and Stones (easy)
- Solution: Jewels and Stones (easy)
- Problem 3: Unique Number of Occurrences (easy)
- Solution: Unique Number of Occurrences (easy)
- Problem 4: Longest Substring Without Repeating Characters (medium)
- Solution: Longest Substring Without Repeating Characters (medium)

10. Heap
9 chapters

- Introduction to Heap
- Take Gifts From the Richest Pile(easy)
- Solution: Take Gifts From the Richest Pile(easy)
- Sort Characters By Frequency (easy)
- Solution: Sort Characters By Frequency (easy)
- Minimum Cost to Connect Sticks(medium)
- Solution: Minimum Cost to Connect Sticks(medium)
- Find the Median of a Number Stream (medium)
- Solution: Find the Median of a Number Stream

11. Graph
11 chapters

- Introduction to Graph
- Graph Representations
- Graph as an Abstract Data Type (ADT)
- Graph Traversal - Depth First Search(DFS)
- Graph Traversal - Breadth First Search (BFS)
- Problem 1: Find if Path Exists in Graph(easy)
- Solution: Find if Path Exists in Graph(easy)
- Problem 2: Number of Provinces (medium)
- Solution: Number of Provinces (medium)
- Problem 3: Minimum Number of Vertices to Reach All Nodes(medium)
- Solution: Minimum Number of Vertices to Reach All Nodes(medium)

12. Trie (aka Prefix Tree)
11 chapters

- Introduction to Trie
- Implement Trie (Prefix Tree) (medium)
- Solution: Implement Trie (Prefix Tree)
- Index Pairs of a String (easy)
- Solution: Index Pairs of a String
- Extra Characters in a String (medium)
- Solution: Extra Characters in a String
- Search Suggestions System (medium)
- Solution: Search Suggestions System
- Design Add and Search Words Data Structure (medium)
- Solution: Design Add and Search Words Data Structure

13. Greedy Algorithm
13 chapters

- Introduction to Greedy Algorithm
- Valid Palindrome II (easy)
- Solution: Valid Palindrome II
- Maximum Length of Pair Chain (medium)
- Solution: Maximum Length of Pair Chain
- Minimum Add to Make Parentheses Valid (medium)
- Solution: Minimum Add to Make Parentheses Valid
- Remove Duplicate Letters (medium)
- Solution: Remove Duplicate Letters
- Largest Palindromic Number (Medium)
- Solution: Largest Palindromic Number
- Removing Minimum and Maximum From Array (medium)
- Solution: Removing Minimum and Maximum From Array

14. Divide and Conquer Algorithm
11 chapters

- Introduction to Divide and Conquer Algorithm
- Majority Element (easy)
- Solution: Majority Element
- Longest Nice Substring (easy)
- Solution: Longest Nice Substring
- Sort List (medium)
- Solution: Sort List
- Beautiful Array (medium)
- Solution: Beautiful Array
- Median of Two Sorted Arrays (medium)
- Solution: Median of Two Sorted Arrays